

Creative Programming: Fundamentals

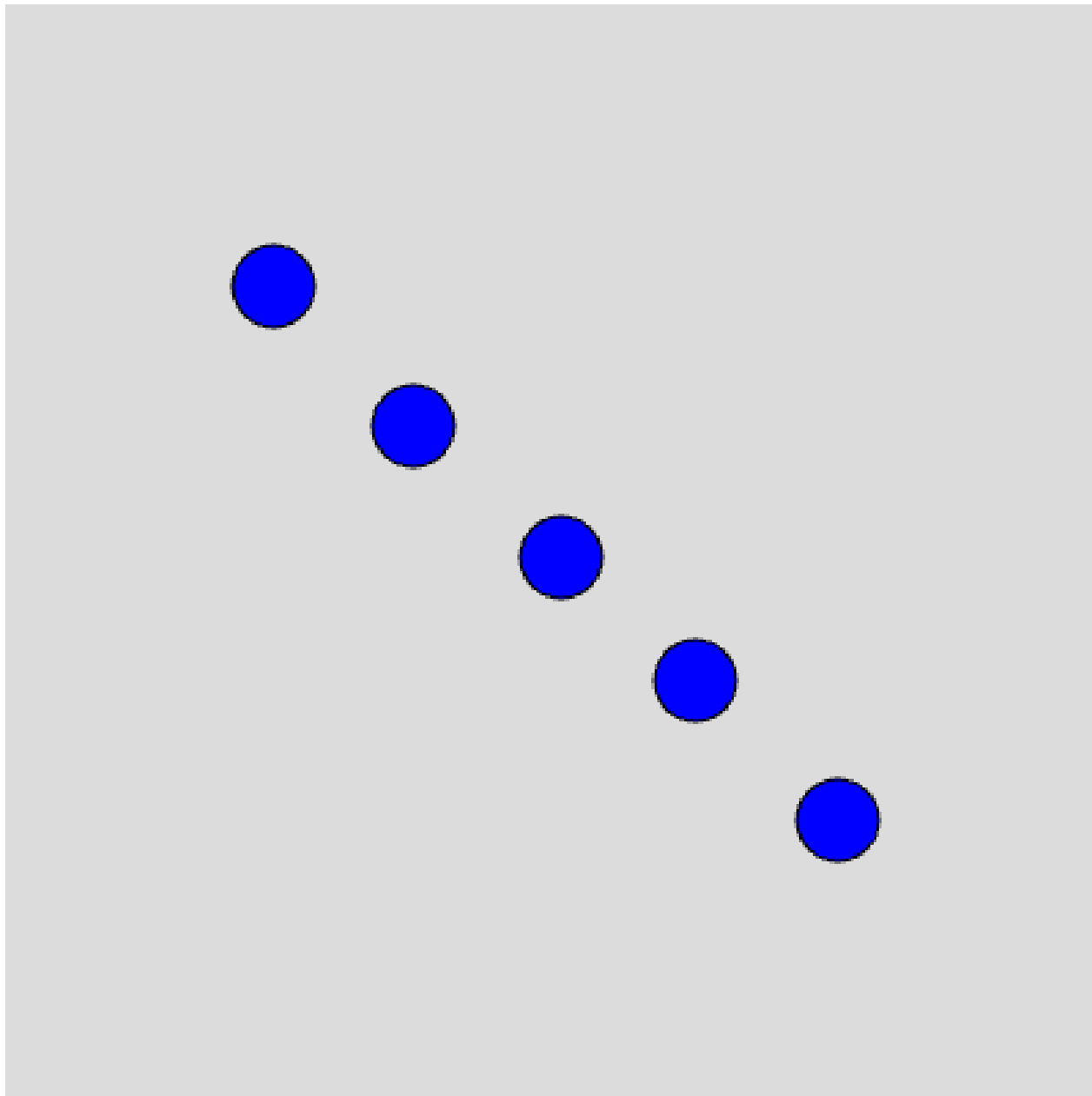
Lecture 2 (3 Feb)

Narration

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- Reference, System Variables, and Recap
- Storing and Retrieving Data
- Logic
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- Arrays

Narration



Narration

(Re)visiting slides?

You can jump to any slide with the menu to the (bottom) left.

Did you miss a slide or want to revisit?

Open the narration tab while studying to get an explanation of difficult slides.



Narration

Handing in exercises

- Programming is all about practice, so that is what the exercises are all about
- You submit on LearnIT the things you want feedback on
- There will be a proper hand-in instruction pdf on LearnIT later today

Narration

Reference, System Variables, and Recap

Narration

Do programmers know everything???

- Programming is about knowing how to express yourself in a programming language ...
- ... but you do not need to remember all functions!

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p5js Reference

- Want to draw a circle, but unsure about the arguments? -> Check the p5js reference
- Don't know how to change the background colour? -> Check the p5js reference

Reference

Find easy explanations for every piece of p5.js code.

Narration

Slides reference

- The slides also contain code examples for many p5js functions
- This week will be very code-heavy, so please refer to the slides and code examples often!
- All code examples are available on GitHub as well. There are over 20 for just this week!

Narration

mouseX and mouseY

- Last week we briefly introduced mouseX and mouseY, but what are they exactly?
- According to the p5js reference:
 - A Number **system variable** that tracks the mouse's horizontal position.

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System Variables

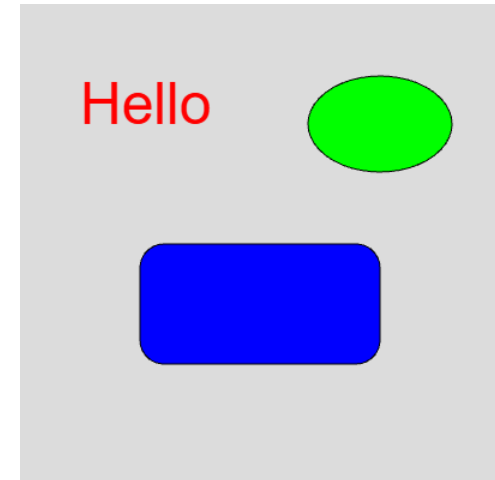
- Named containers that the system manages for us
- Created and managed automatically by p5js
- Examples:
 - **mouseX**: horizontal position of the mouse
 - **mouseY**: vertical position of the mouse
 - **width**: horizontal size of the canvas
 - **height**: vertical size of the canvas
 - **frameCount**: amount of times draw() has been called since start

Narration

Recap: Shapes

- `text()`, `ellipse()`, and `rect()`
- You need to specify the coordinates (`x`, `y`) and the size / contents

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3   background(220); // grey background  
4  
5   // Set text properties  
6   textSize(48); // big text  
7   fill(255, 0, 0); // red colour  
8   text("Hello", 50, 100);  
9  
10  // Draw an ellipse  
11  fill(0, 255, 0); // green colour  
12  ellipse(300, 100, 120, 80);  
13  
14  // Draw a rounded rectangle  
15  fill(0, 0, 255); // blue colour  
16  rect(100, 200, 200, 100, 20); // The 5th argument is *optional* and defines the corner
```



Narration

Mathing out

- We can use operators such as `+`, `-`, `*`, and `/`.
- This allows you to e.g. add an offset to the mouse position
- `+` can also add strings together

```
1 ▾ function setup() {  
2   createCanvas(500, 500);  
3   textSize(48);  
4 }  
5  
6 ▾ function draw() {  
7   background(220);  
8  
9   fill(100, 200, 255); // rect colour  
10  // x is 25 pixels to the right of the mouse  
11  // y is centered vertically  
12  // we use the 25 to center the rectangle due to its 50 pixel width and height  
13  rect(mouseX + 50 - 25, height / 2 - 25, 50, 50);
```

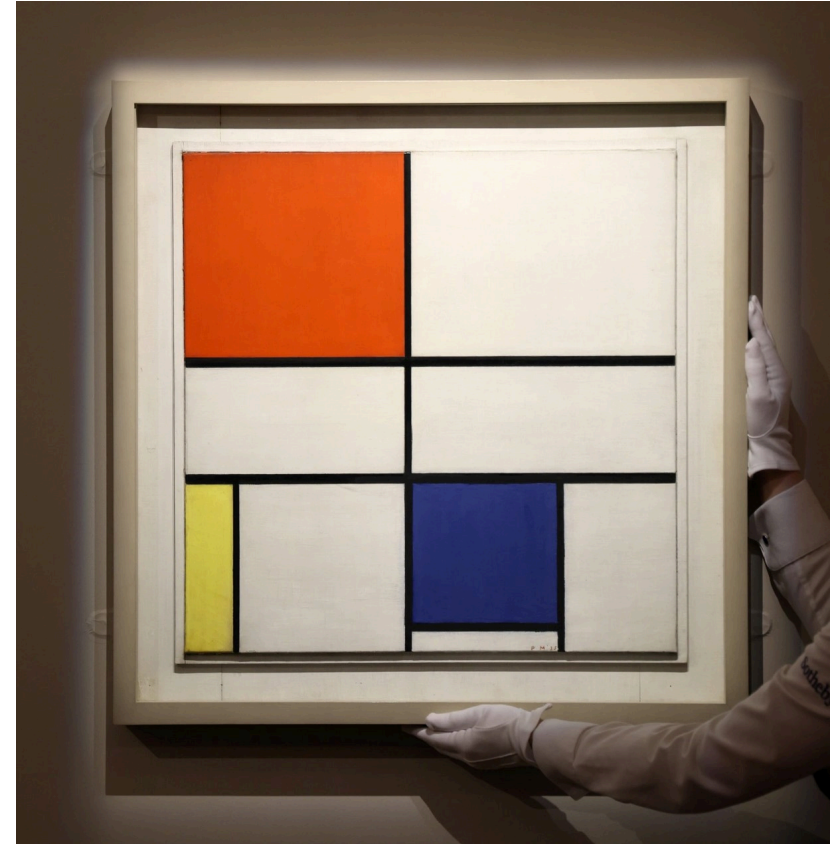
x: 0 offset: 50
y: 0



divided: 0
frameCount: 73

Narration

Mondrian

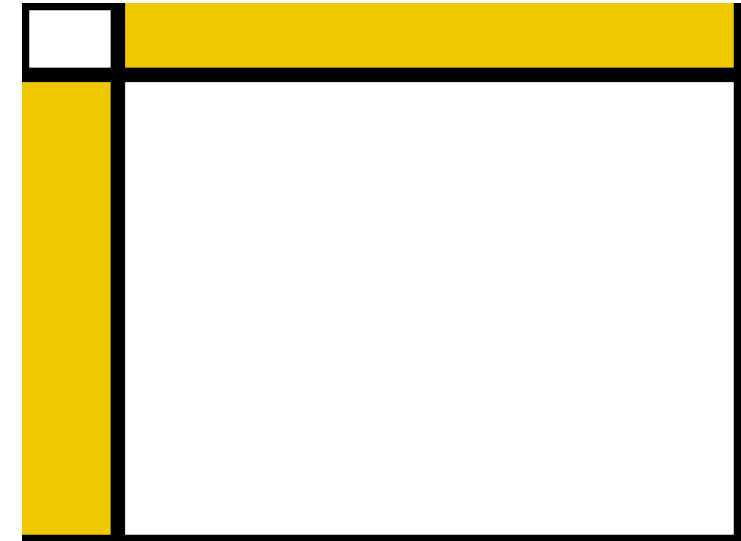


Narration

System Variables + Shapes

- With a bit of creativity we can make an interactive ‘Mondrian’ painting!

```
1 ▾ function setup() {  
2   createCanvas(600, 450);  
3 }  
4  
5 ▾ function draw() {  
6   stroke(0);  
7   strokeWeight(12);  
8  
9   // Top-Left  
10  fill(255, 255, 255);  
11  rect(0, 0, width, height);  
12  
13  // Top-middle  
14  fill(0,0,255);  
15  rect(mouseX - 80, 0, 160, mouseY - 60);  
16  
17  // Middle-Left  
18  fill(255,0,0);
```



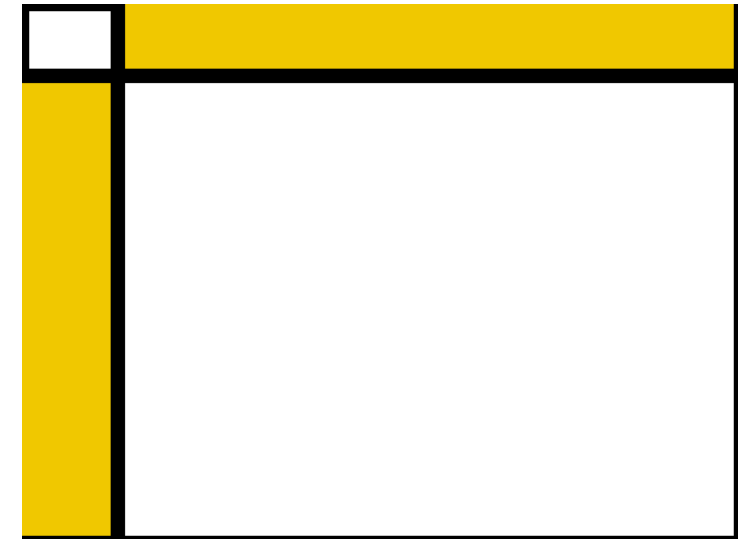
Narration

Your own Mondrian

💡 Class Assignment

Create your own interactive Mondrian painting! In groups of 1-3, make a p5js sketch that draws rectangles (or other shapes) based on the mouse position. Be creative with colors and shapes!

```
1 ▾ function setup() {  
2   createCanvas(600, 450);  
3 }  
4  
5 ▾ function draw() {  
6   stroke(0);  
7   strokeWeight(12);  
8  
9   // Top-Left  
10  fill(255, 255, 255);  
11  rect(0, 0, width, height);  
12  
13  // Top-middle  
14  fill(0,0,255);
```



Narration

Storing and Retrieving Data

Narration

Exercise 1: Routine

- Going back to the first step of exercise 1:
 - Write down a daily routine, step by step
 - You go through these steps every day, in the same order
- How can we apply this to programming?

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Imperative programming

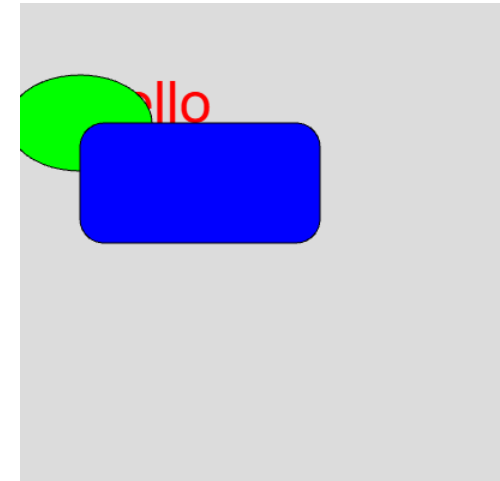
- Programming is like a routine
- You tell the computer what to do, step by step
- The computer evaluates the code line by line, top to bottom

Narration

Overlapping

- When drawing, new shapes go over old shapes
- The order of shapes matter

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3   background(220); // grey background  
4  
5   // Set text properties  
6   textSize(48); // big text  
7   fill(255, 0, 0); // red colour  
8   text("Hello", 50, 100);  
9  
10  // Draw an ellipse  
11  fill(0, 255, 0); // green colour  
12  ellipse(50, 100, 120, 80);  
13  
14  // Draw a rounded rectangle  
15  fill(0, 0, 255); // blue colour  
16  rect(50, 100, 200, 100, 20); // The 5th argument is *optional* and defines the corner
```

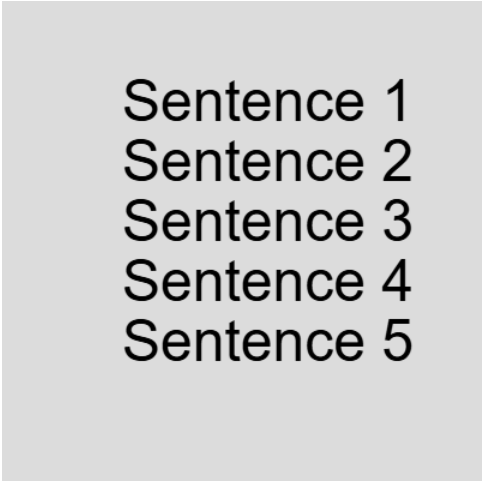


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Drawing multiple lines of text

- Per line we can use a `text()` call
- We need to manually space these 50 pixels vertically apart

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3     background(220); // grey background  
4  
5     // Set text properties  
6     textSize(48); // big text  
7  
8     text("Sentence 1", 100, 100);  
9     text("Sentence 2", 100, 150);  
10    text("Sentence 3", 100, 200);  
11    text("Sentence 4", 100, 250);  
12    text("Sentence 5", 100, 300);  
13 }  
14  
15 ▾ function draw() {  
16    // No animation needed for this demonstration
```



Sentence 1
Sentence 2
Sentence 3
Sentence 4
Sentence 5

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Memorizing

- What if we want to draw more lines?
- We would need to write more `text()` calls and calculate the `y` position each time
- This is tedious and error-prone

Narration

Variables

- A **variable** is a named container for data

```
1 let interestingNumber = 42;
```

- Here we have used `let` to create a variable named `interestingNumber` that contains the number `42`
- We can now use `interestingNumber` in our code instead of having to write `42` every time
- You can store anything in a variable: numbers, text, etc.

Narration

Reassigning variables

- We can change the contents of a variable at any time
- This is called **reassigning** the variable
- We only use `let` once, when we first create the variable

```
1 ▾ function setup() {  
2   createCanvas(400, 200);  
3   background(240);  
4  
5   // Start with interestingNumber set to 42  
6   let interestingNumber = 42;  
7  
8   // Draw the initial value as text  
9   textSize(36);  
10  text('interestingNumber: ' + interestingNumber, 20, 60);  
11  
12  // Reassign to a new value  
13  interestingNumber = 100; // reassigned
```

interestingNumber: 42
reassigned to: 100

Narration

Reassigning variables

- `interestingNumber = 100;` reassigns the variable to now contain `100`
- `interestingNumber++;` increases the variable by `1` (only numbers)
- `interestingNumber += 5;` increases the variable by `5`

Narration

Data Types: String

- We have seen the **string** data type before
- Strings are sequences of characters, enclosed in quotes ' ' or ""

```
1 let greeting = "Hello, world!";
```

Narration

Data Types: Integer

- An **integer** is a whole number (no decimal point)
- They can be positive, negative, or 0
- You can convert other types to integers using `int()`

```
1 let wholeNumber = 42;  
2 let convertedNumber = int("123"); // converts string to integer
```

Narration

Data Types: Float

- A **float** (floating-point number) is a number with a decimal point
- They can also be positive, negative, or 0
- You can convert other types to floats using `float()`
- `round()`, `floor()`, and `ceil()` can round, round down, and round up floats to integers, respectively

```
1 let decimalNumber = 3.14;  
2 let convertedFloat = float("3.14"); // converts string to float  
3 let roundedNumber = round(3.7); // rounds to 4
```

[Narration](#)

Data Types: Boolean

- A **boolean** represents a logical value: `true` or `false`
- `true` and `false` are system keywords, similar to e.g. `mouseX` and `width`
- Booleans are often used in conditional statements to control the flow of a program

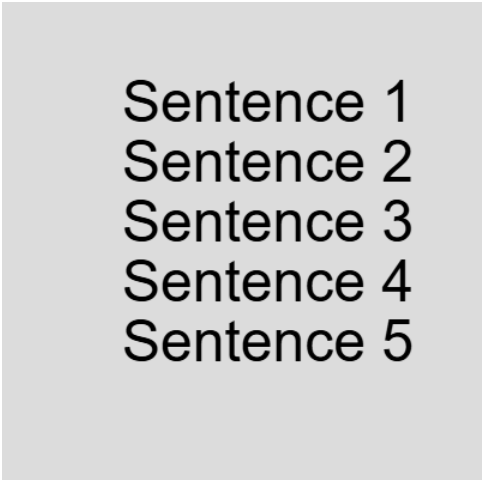
```
1 let isRaining = true;  
2 let isSunny = false;
```

Narration

Using a variable for y position

- We set `verticalPosition` to `100` at the start
- Each time we draw a line of text, we increase `verticalPosition` by `50`
- If we want more lines we can copy-paste the two lines

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3     background(220); // grey background  
4  
5     // Set text properties  
6     textSize(48); // big text  
7  
8     let verticalPosition = 100;  
9     text("Sentence 1", 100, verticalPosition);  
10  
11     verticalPosition += 50;  
12     text("Sentence 2", 100, verticalPosition);  
13
```



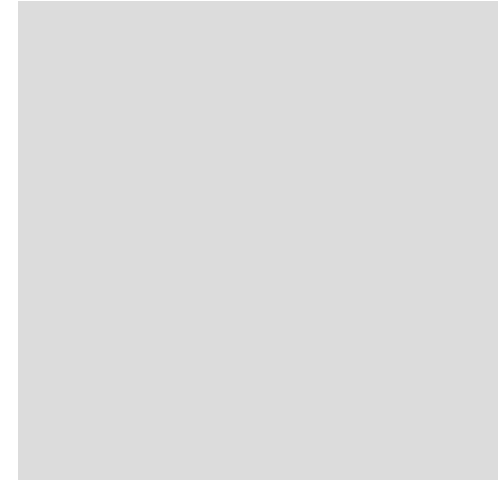
Sentence 1
Sentence 2
Sentence 3
Sentence 4
Sentence 5

Narration

Building the example: variable ball

- Often we want to track the position of something
- Variables allow us to store and update this position over time

```
1  let x = 0;
2
3  ▾ function setup() {
4      createCanvas(400, 400);
5  }
6
7  ▾ function draw() {
8      background(220);
9
10     // Draw a circle at position x
11     fill(255, 0, 0);
12     ellipse(x, 200, 50, 50);
13
14     // Move the circle
15     x += 2;
16
```



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Logic

Narration

Conditional Statements

- Sometimes we want to only do something if a certain condition is met
 - E.g. reset the ball when it leaves the screen
- This is done with **conditional statements**

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Conditions

- A **condition** is an expression that is `true` or `false` (boolean)
- We can use comparison operators to create conditions:
 - `==` (equal to)
 - `!=` (not equal to)
 - `<` (less than)
 - `>` (greater than)
 - `<=` (less than or equal to)
 - `>=` (greater than or equal to)

Narration

If statement

- An **if statement** checks a condition
- If the condition is true, the code inside the if statement is executed
- In the example below, we check whether `mouseX` is smaller than `200`.

```
1 function draw() {  
2   background(220);  
3   fill(0, 0, 255); // blue  
4  
5   // Change color based on mouse position  
6   if (mouseX < 200) {  
7     fill(255, 0, 0); // red  
8   }  
9  
10  ellipse(mouseX, mouseY, 50, 50);  
11 }
```

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If statement

- The evaluated condition is between parentheses ()
- The code that is executed on true is between brackets { }
- Indentation is not required, but makes code easier to read

```
1 function draw() {  
2     background(220);  
3     fill(0, 0, 255); // blue  
4  
5     // Change color based on mouse position  
6     if (mouseX < 200) {  
7         fill(255, 0, 0); // red  
8     }  
9  
10    ellipse(mouseX, mouseY, 50, 50);  
11 }
```

Narration

If-else statement

- What happens if the condition is false?
- We can use an **if-else statement** to provide alternative code to execute
- True -> do the first thing. False -> do the second thing.

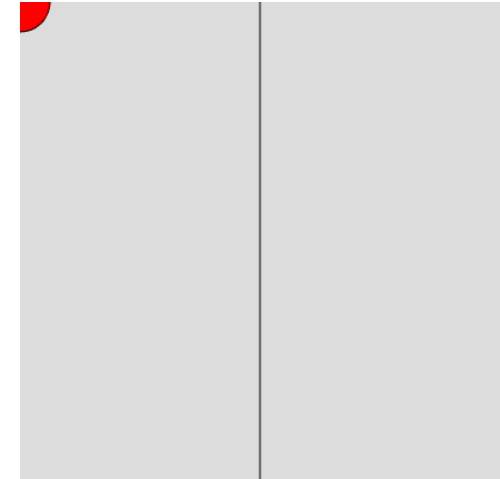
```
1 function draw() {  
2     background(220);  
3  
4     // Change color based on mouse position  
5     if (mouseX < 200) {  
6         fill(255, 0, 0); // red  
7     } else {  
8         fill(0, 0, 255); // blue  
9     }  
10  
11     ellipse(mouseX, mouseY, 50, 50);  
12 }
```

[Narration](#)

Example: mouse circle

- Change the circle color based on mouse position
- If `mouseX < 200` -> red
- Else -> blue

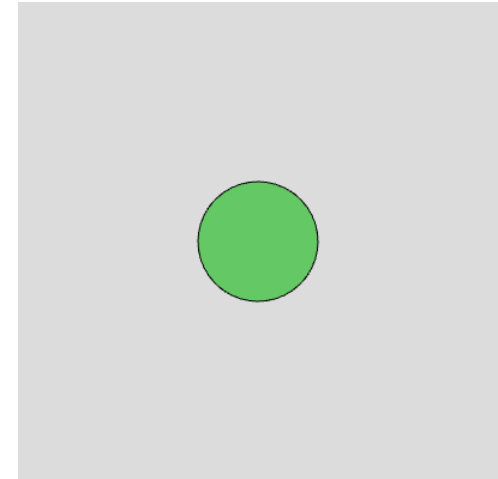
```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6     background(220);  
7  
8     // Draw a line dividing the canvas  
9     stroke(0);  
10    line(200, 0, 200, 400);  
11  
12    // Change color based on mouse position  
13 ▾ if (mouseX < 200) {
```

[Narration](#)

Example: two shapes

- Draw a shape based on vertical mouse position
- If `mouseY < 200` -> draw ellipse
- Else -> draw rectangle

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6     background(220);  
7  
8     fill(100, 200, 100);  
9  
10    // Draw different shapes based on mouse y position  
11 ▾    if (mouseY < 200) {  
12        ellipse(200, 200, 100, 100);  
13 ▾    } else {
```



Narration

Nesting

- We can use if statements inside other if statements: **nesting**
- This allows us to check multiple conditions in a structured way

```
1 ▾ function setup() {  
2     createCanvas(600, 600);  
3 }  
4  
5 ▾ function draw() {  
6     background(220);  
7     // Nested if: decide which quadrant the mouse is in  
8     textSize(40);  
9     fill(0);  
10  
11 ▾ if (mouseX < width / 2) {  
12     // Left half  
13 ▾ if (mouseY < height / 2) {  
14     // Top-left  
15     text('Top-left', 20, 30);  
16     fill(180, 220, 180);
```

Top-left



Narration

Conditional operators

- We can combine conditions using **conditional operators**
- Common operators:
 - `&&` (and): both conditions must be true
 - `||` (or): at least one condition must be true

Narration

If else if

- We can chain multiple conditions using **if else if** statements
- This allows us to check several conditions in sequence

```
1 ▾ function setup() {  
2     createCanvas(600, 600);  
3 }  
4  
5 ▾ function draw() {  
6     background(240);  
7  
8     // Show current mouse coordinates  
9     fill(0);  
10    textSize(32);  
11    text('mouseX: ' + mouseX, 10, 50);  
12    text('mouseY: ' + mouseY, 10, 100);  
13  
14 ▾    if ((mouseX > width / 2) && (mouseY > height / 2)) {  
15        // If mouse bottom right  
16        // AND true -> green circle
```

mouseX: 0
mouseY: 0



Narration

Storing conditions

- We can store the result of a condition in a variable
- This can make code easier to read and manage

```

1  ▾ function setup() {
2      createCanvas(600, 600);
3  }
4
5  ▾ function draw() {
6      background(240);
7
8      // Show current mouse coordinates
9      fill(0);
10     textSize(32);
11     text('mouseX: ' + mouseX, 10, 50);
12     text('mouseY: ' + mouseY, 10, 100);
13
14     // Store conditions in variables
15     let inRightHalf = mouseX > width / 2;
16     let inBottomHalf = mouseY > height / 2;

```

mouseX: 0
mouseY: 0

inRightHalf: false
inBottomHalf: false



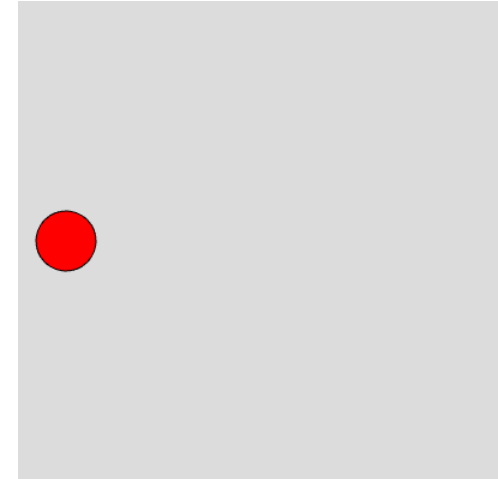
AND (inRightHalf && inBottomHalf): false
OR (inRightHalf || inBottomHalf): false

Narration

Building the example: resetting ball

- We can use conditions to check if the ball goes offscreen
- If it does, we reset its position

```
1  let x = 0;
2
3  ▾ function setup() {
4      createCanvas(400, 400);
5  }
6
7  ▾ function draw() {
8      background(220);
9
10     // Draw a circle at position x
11     fill(255, 0, 0);
12     ellipse(x, 200, 50, 50);
13
14     // Move the circle
15     x += 2;
16
```

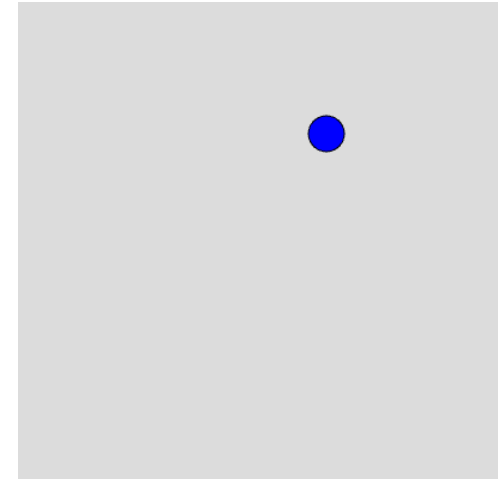


Narration

Building the example: bouncing ball

- With variables and conditions, we can make the ball bounce off the edges
- Instead of resetting the position, reverse the direction when hitting an edge

```
1  let x = 200;
2  let y = 200;
3  let speedX = 3;
4  let speedY = 2;
5  let diameter = 30;
6
7  function setup() {
8    createCanvas(400, 400);
9  }
10
11 function draw() {
12   background(220);
13
14   // Draw the ball
```



Narration

Looping

Narration

Repeating lines of code

- Remember our text example; we want to draw many lines of text
- Copy-pasting code is tedious and error-prone
- We can use **loops** to repeat code multiple times

Narration

For loop

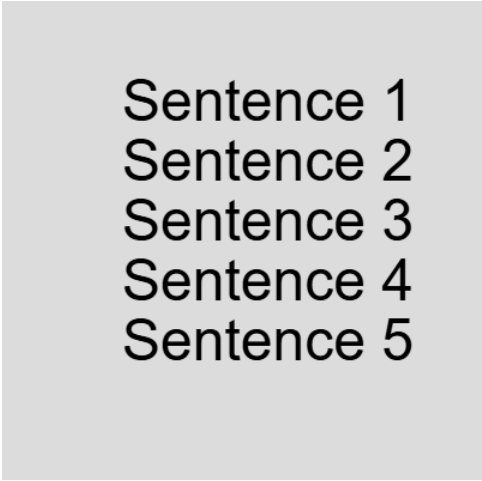
- A **for loop** repeats a block of code a specified number of times
- `for (initialization; condition; update) { // code to repeat }`
 - **initialization**: set up a variable to track iterations
 - **condition**: when to stop looping
 - **update**: change the variable each iteration

Narration

Lines example

- By putting our text drawing code inside a for loop, we can draw many lines easily
- In this example, we loop with variable `i` from `1` to `5`

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3     background(220); // grey background  
4  
5     // Set text properties  
6     textSize(48); // big text  
7  
8     let verticalPosition = 100;  
9  
10 ▾ for(let i = 1; i <= 5; i++) { // Loop from 1 to 5  
11     text("Sentence " + i, 100, verticalPosition);  
12     verticalPosition += 50;  
13 }  
14 }
```



Sentence 1
Sentence 2
Sentence 3
Sentence 4
Sentence 5

Narration

Looping with different update

- We can also use a different update step so that we do not need 2 variables

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3     background(220); // grey background  
4  
5     // Set text properties  
6     textSize(48); // big text  
7  
8 ▾   for(let verticalPosition = 100; verticalPosition <= 350; verticalPosition += 50) { // Loop  
    from 1 to 5  
9       text("Sentence " + verticalPosition, 100, verticalPosition);  
10    }  
11 }  
12  
13 ▾ function draw() {  
14     // No animation needed for this demonstration  
15 }  
16
```

Sentence 100
Sentence 150
Sentence 200
Sentence 250
Sentence 300
Sentence 350

Narration

Scope

- **Scope:** Variables defined inside a loop are only accessible inside that loop
- Similarly, variables defined in `draw()` are not accessible in `setup()`
- You can define variables outside of functions to give them a broader scope

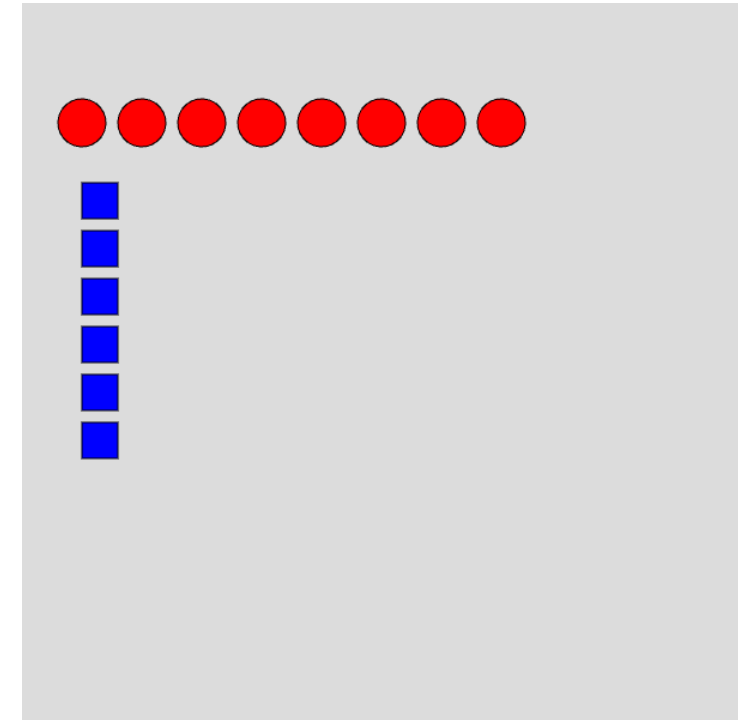
```
1 let firstVar = 5; // global (everywhere) scope
2
3 function setup() {
4     let secondVar = 10; // local to setup
5 }
6
7 function draw() {
8     let thirdVar = 20; // local to draw
9 }
```

[Narration](#)

Looping and scope

- In this example, the variable `i` is contained within the loop
- We can thus define a new variable `i` elsewhere without conflict

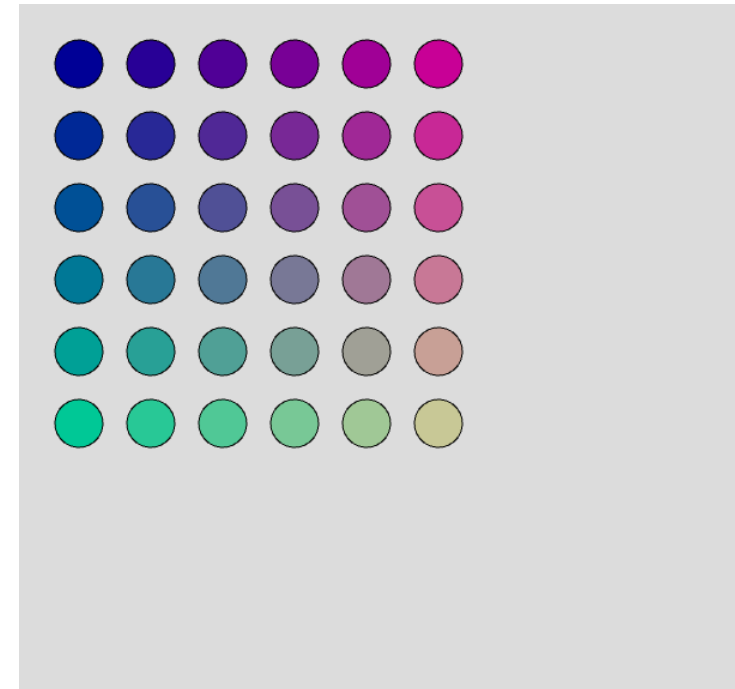
```
1 ▾ function setup() {  
2     createCanvas(600, 600);  
3     background(220);  
4  
5     // Draw a row of circles using a loop  
6 ▾ for (let i = 0; i < 8; i++) {  
7         fill(255, 0, 0);  
8         ellipse(50 + i * 50, 100, 40, 40);  
9     }  
10  
11     // Draw a column of squares using a loop  
12 ▾ for (let i = 0; i < 6; i++) {  
13         fill(0, 0, 255);  
14         rect(50, 150 + i * 40, 30, 30);  
15     }  
16 }
```

[Narration](#)

Nesting loops

- Similar to if statements, we can nest loops inside other loops
- This allows us to create more complex patterns and structures, usually 2D ones

```
1 ▾ function setup() {  
2   createCanvas(600, 600);  
3   background(220);  
4  
5   // Draw a grid of circles using nested loops  
6 ▾   for (let i = 0; i < 6; i++) {  
7 ▾     for (let j = 0; j < 6; j++) {  
8       fill(i * 40, j * 40, 150);  
9       ellipse(50 + i * 60, 50 + j * 60, 40, 40);  
10      }  
11    }  
12  }  
13  
14 ▾ function draw() {
```

[Narration](#)

Arrays

Narration

Storing multiple values

- Sometimes we want to store multiple values
- We can use an **array** to store a list of values in a single variable
- An array is created using square brackets `[]`, with values separated by commas `,`

```
1 let colours = ['red', 'green', 'blue'];
```

- Here, `colours` is an array containing three strings: 'red', 'green', and 'blue'

Narration

Accessing array elements

- We can access individual elements in an array using their index
- Array indices start at 0
- To access the first element, we use `arrayName[0]`, the second element is `arrayName[1]`, and so on

```
1 let colours = ['red', 'green', 'blue'];  
2 let firstColour = colours[0]; // 'red'  
3 let secondColour = colours[1]; // 'green'
```

[Narration](#)

Arrays of text

- Revisiting our text lines example, we can store the lines in an array
- This allows us to easily manage and access the content of the lines

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3     background(220); // grey background  
4  
5     // Set text properties  
6     textSize(48); // big text  
7  
8     // Define an array of strings  
9     let sentences = ["Sentence 1", "Sentence 2", "Sentence 3", "Sentence 4", "Sentence 5"];  
10  
11     // Access array elements by index  
12     text(sentences[0], 100, 100);  
13     text(sentences[1], 100, 150);  
14     text(sentences[2], 100, 200);  
15     text(sentences[3], 100, 250);  
16     text(sentences[4], 100, 300);
```

Sentence 1
Sentence 2
Sentence 3
Sentence 4
Sentence 5

Narration

Using arrays

- Arrays have some useful properties and methods, such as
 - `myArray.length` gives the number of elements in the array
 - `myArray.push(value)` adds a new value to the end of the array
 - `myArray.pop()` removes the last value from the array
 - `myArray.sort()` sorts the array elements
 - `myArray.reverse()` reverses the order of the array elements
- Find more in the p5js reference

Narration

For loop + array

- Arrays and loops work well together
- We can use a for loop to iterate over the elements of an array

```
1 ▾ function setup() {  
2     createCanvas(400, 400);  
3     background(220); // grey background  
4  
5     // Set text properties  
6     textSize(48); // big text  
7  
8     // Define an array of strings  
9     let sentences = ["Sentence 1", "Sentence 2", "Sentence 3", "Sentence 4", "Sentence 5"];  
10  
11     // Use a for loop to iterate through the array  
12 ▾ for (let i = 0; i < sentences.length; i++) {  
13         text(sentences[i], 100, 100 + i * 50);  
14     }  
15 }  
16
```

Sentence 1
Sentence 2
Sentence 3
Sentence 4
Sentence 5

Narration

Nesting arrays

- Arrays can also contain other arrays, creating multi-dimensional arrays
- This is useful for storing grid-like data structures

```
1 // Define a 2D array (grid)
2 ▾ let grid = [
3     [1, 2, 3],
4     [4, 5, 6],
5     [7, 8, 9]
6 ];
7
8 ▾ function setup() {
9     createCanvas(300, 300);
10 }
11
12 ▾ function draw() {
13     background(220);
14     fill(0);
15     textSize(32);
16 }
```

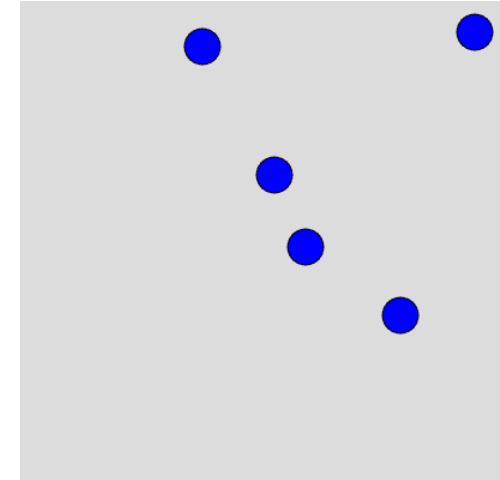
1	2	3
4	5	6
7	8	9

[Narration](#)

Building the example: bouncing ball

- Using arrays we can define multiple balls
- We can use a for loop to update and draw all balls

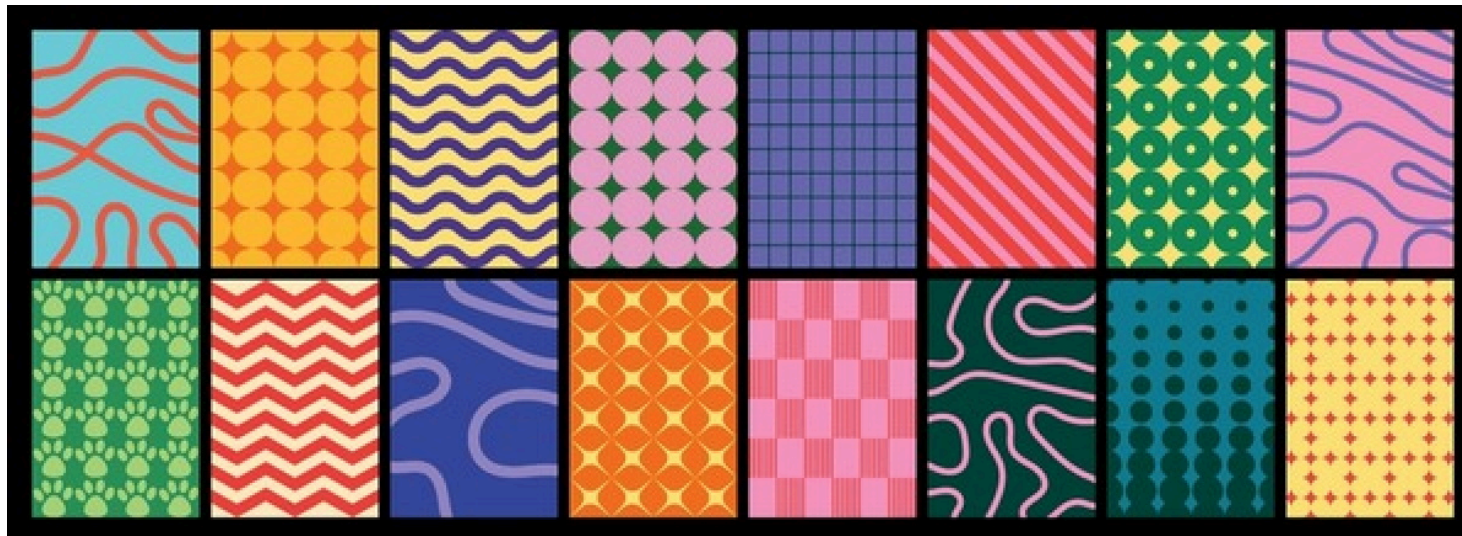
```
1 let x = [200, 100, 300, 150, 250];
2 let y = [200, 100, 300, 150, 250];
3 let speedX = [3, -2, 4, -1, 2];
4 let speedY = [2, 3, -2, 4, -3];
5 let diameter = 30;
6
7 function setup() {
8   createCanvas(400, 400);
9 }
10
11 function draw() {
12   background(220);
13
14   // Go over every ball
15   for (let i = 0; i < 5; i++) {
16     // Draw the ball
```

[Narration](#)

Put arrays and loops to the test!

💡 Class Assignment: Patterns!

1. In a group of 1-3, start by drawing some patterns on paper or a whiteboard (e.g. <https://www.tldraw.com>)
2. Then, implement these patterns in p5js using loops and arrays where appropriate.
3. Can you make them interactive using mouseX and mouseY?



Narration

Summary

- We covered variables, data types, conditional statements, loops, and arrays
- These are fundamental programming concepts that will help you create more complex and interactive sketches
- Practice these concepts; it is a lot for one week, do not be afraid to make mistakes.
- Project groups will be published today.

Narration